HOW THE CONFORMAL MODULUS OF AN ANNULUS QUANTITATIVELY CONTROLS HOLOMORPHIC MAPS

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ABSTRACT. Every nondegenerate doubly connected planar domain is conformally equivalent to $A(r,R)$ [3, 9], where

$$m := \operatorname{mod}(A) = \frac{1}{2\pi}\log\frac{R}{r}.$$

We identify m with radial extremal length and reciprocal capacity [2, 15, 20]. We prove sharp centered Fourier–Laurent damping [7, 19], establish Schwarz–Pick estimates [6, 5, 12], and classify proper maps [9, 7]. We also describe boundary transfer and degeneration and quantify quasiconformal modulus distortion [17, 13, 22]. In round coordinates, a proper map of signed degree $n \in \mathbb{Z}\setminus\{0\}$ has

$$f(z) = c\cdot z^n, \qquad c\in\mathbb{C}\setminus\{0\}, \qquad \operatorname{mod}(A') = |n|\cdot\operatorname{mod}(A).$$

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Part 1. Modulus Control on Annuli

1. Introduction

Throughout, A is a nondegenerate doubly connected planar domain, $\operatorname{mod}(A)$ is its conformal modulus, and every hyperbolic metric has curvature -1; the classical background is contained in [3, 7, 9]. Every such domain has a round model determined by one positive parameter.

$$A \cong A(r,R) := \{z \in \mathbb{C} : r < |z| < R\}, \qquad m := \operatorname{mod}(A) = \frac{1}{2\pi}\log\left(\frac{R}{r}\right),$$

$$q := \frac{r}{R} = e^{-2\pi m}, \qquad A(r,R) \cong A(q,1), \qquad W := \log\left(\frac{R}{r}\right) = 2\pi m.$$

The logarithmic cover turns m into the Euclidean width W.

$$\begin{tikzcd} S_W \arrow[r, "w\mapsto e^w"] \arrow[d, two heads, "w\sim w+2\pi i"'] & A(q,1) \\ S_W/(2\pi i\mathbb{Z}) \arrow[ur, "\cong"'] & \end{tikzcd} \qquad S_W := \{w \in \mathbb{C} : -W < \Re w < 0\}.$$

Variational geometry, periodic analysis, and covering equivariance consequently depend on the same parameter. Proper maps exhibit exact rigidity, whereas arbitrary holomorphic maps satisfy intrinsic contraction and mode-damping estimates rather than a monomial classification.

1.1. Result.

The principal identities separate these geometric, analytic, and topological forms of control.

Theorem A. *Let $A = A(r,R)$, $m = \operatorname{mod}(A)$, and $W = 2\pi m$. Then*

(1) *For the connecting and separating families Γ_{rad} and Γ_{sep},*
$$\operatorname{Ext}_A(\Gamma_{\text{rad}}) = m, \qquad \operatorname{Ext}_A(\Gamma_{\text{sep}}) = m^{-1}, \qquad \operatorname{cap}(A) = m^{-1}.$$

(2) *For $C_* := \{|z| = \sqrt{rR}\}$,*
$$\rho_A(z) = \frac{\pi}{W|z|}\csc\left(\frac{\pi}{W}\log\left(\frac{|z|}{r}\right)\right), \qquad \ell_A(C_*) = \frac{\pi}{m}.$$

(3) *If $f \in \operatorname{Hol}(A,\mathbb{D})$ and $\rho_{\mathbb{D}}(\zeta) := 2(1-|\zeta|^2)^{-1}$, then*
$$\rho_{\mathbb{D}}(f(z))\cdot|f'(z)| \le \rho_A(z), \qquad d_{\mathbb{D}}(f(z),f(\zeta)) \le d_A(z,\zeta).$$

(4) *If $A = A(q,1)$, $f(z) = \sum_{n\in\mathbb{Z}} a_n z^n$, and $\sup_{A(q,1)}|f| \le M$, then*
$$|a_n| \le M \quad (n \ge 0), \qquad |a_{-n}| \le Mq^n \quad (n \ge 1), \qquad \sup_{|z|=\sqrt{q}} |f(z) - a_0| \le \frac{2M\sqrt{q}}{1-\sqrt{q}}.$$

(5) *Let $A' = A(r',R')$. If $g \in \operatorname{Hol}(A,A')$ is proper of degree $n \in \mathbb{Z}\setminus\{0\}$, then, in round coordinates,*
$$g(z) = c\cdot z^n, \qquad c \in \mathbb{C}\setminus\{0\}, \qquad \operatorname{mod}(A') = |n|\cdot\operatorname{mod}(A).$$

(6) *If $h\colon A \to A'$ is a K-quasiconformal homeomorphism, then*
$$K^{-1}\cdot\operatorname{mod}(A) \le \operatorname{mod}(A') \le K\cdot\operatorname{mod}(A).$$

Proof. § 9. □

2. Annuli and conformal modulus

The Riemann mapping theorem leaves no continuous conformal parameter for proper simply connected planar domains, whereas doubly connected domains retain one [1, 7, 9].

2.1. **Logarithmic coordinates.**

Definition 2.1.1 (Nondegenerate annulus, [3, 9])**.** A domain $A \subset \mathbb{C}$ is an *annulus* if $\widehat{\mathbb{C}} \setminus A$ has two connected components, and is *nondegenerate* if neither component is a point. For $0 < r < R$, write

$$A(r,R) := \{z \in \mathbb{C} : r < |z| < R\}.$$

Theorem 2.1.2 (Annulus uniformization, [3, 9, 15, 16])**.** *Every nondegenerate annulus is biholomorphic to $A(q,1)$ for a unique $q \in (0,1)$.*

Definition 2.1.3 (Conformal modulus, [1, 15, 22])**.** If $A \cong A(q,1)$ as in Theorem 2.1.2, define

$$\operatorname{mod}(A) := \frac{1}{2\pi} \log\left(\frac{1}{q}\right), \qquad q = e^{-2\pi \operatorname{mod}(A)}.$$

In particular,

$$A(r,R) \cong A(r/R, 1), \qquad \operatorname{mod}(A(r,R)) = \frac{1}{2\pi} \log\left(\frac{R}{r}\right).$$

Proposition 2.1.4 (Elementary normalizations, [1, 14])**.** *For $\lambda \in \mathbb{C} \setminus \{0\}$, scaling and inversion give*

$$z \mapsto \lambda z \colon A(r,R) \xrightarrow{\cong} A(|\lambda| r, |\lambda| R), \qquad z \mapsto z^{-1} \colon A(r,R) \xrightarrow{\cong} A(R^{-1}, r^{-1}),$$

and both maps preserve modulus.

Proof. The radial inequalities determine both images.

$$r < |z| < R \iff |\lambda| r < |\lambda z| < |\lambda| R, \qquad r < |z| < R \iff R^{-1} < |z^{-1}| < r^{-1}.$$

The maps have holomorphic inverses, and

$$\frac{|\lambda| R}{|\lambda| r} = \frac{1/r}{1/R} = \frac{R}{r} \overset{2.1.3}{\Longrightarrow} \operatorname{mod}(A(|\lambda| r, |\lambda| R)) = \operatorname{mod}(A(R^{-1}, r^{-1})) = \operatorname{mod}(A(r,R)).$$

□

Corollary 2.1.5 (Complete conformal invariant, [3, 16])**.** *For nondegenerate annuli A and B,*

$$A \cong B \iff \operatorname{mod}(A) = \operatorname{mod}(B).$$

Proof. Let $q_A, q_B \in (0,1)$ be supplied by Theorem 2.1.2. Then

$$A \cong B \overset{2.1.2}{\iff} q_A = q_B \iff \log\left(\frac{1}{q_A}\right) = \log\left(\frac{1}{q_B}\right) \overset{2.1.3}{\iff} \operatorname{mod}(A) = \operatorname{mod}(B).$$

□

Fix $0 < r < R$, and put

$$S_{r,R} := \{w \in \mathbb{C} : \log r < \Re w < \log R\}, \qquad W := \log\left(\frac{R}{r}\right).$$

Proposition 2.1.6 (Logarithmic cover, [3, 7, 9])**.** *The map $p(w) := e^w$ is a universal covering $p \colon S_{r,R} \to A(r,R)$, with*

$$\operatorname{Deck}(p) = \{T_k : k \in \mathbb{Z}\}, \qquad T_k(w) := w + 2\pi i k, \qquad A(r,R) \cong S_{r,R}/(2\pi i \mathbb{Z}).$$

Its strip width is $W = 2\pi \operatorname{mod}(A(r,R))$.

Proof. Write $w = u + iv$ and $z = \rho e^{i\theta}$. Then

$$w \in S_{r,R} \implies r < e^u = |e^w| < R, \qquad z \in A(r,R) \implies \log\rho + i\theta \in S_{r,R}, \quad e^{\log\rho + i\theta} = z.$$

Moreover,

$$p'(w) = e^w \neq 0, \qquad p(w_1) = p(w_2) \iff w_1 - w_2 \in 2\pi i\mathbb{Z}.$$

Each sufficiently small disc $U \Subset A(r,R)$ admits a logarithm branch L, and the disjoint sets $L(U) + 2\pi ik$ are the inverse sheets over U. Thus p is a covering; since $S_{r,R}$ is convex, it is universal, and the fiber identity gives $\mathrm{Deck}(p) = \{T_k : k \in \mathbb{Z}\}$. Finally,

$$W = \log(R/r) \overset{2.1.3}{\implies} W = 2\pi \mathrm{mod}(A(r,R)).$$

$\square$

Corollary 2.1.7 (Cylinder model, [1, 9])**.** *For*

$$\mathscr{R}_{r,R} := \{w \in \mathbb{C} : \log r < \Re w < \log R,\ 0 \leq \Im w \leq 2\pi\},$$

the horizontal sides are identified and

$$A(r,R) \cong \mathscr{R}_{r,R}/\big((u,0) \sim (u,2\pi)\big), \qquad \frac{W}{2\pi} = \mathrm{mod}(A(r,R)).$$

Proof.

$$p(w + 2\pi i) = p(w) \overset{2.1.6}{\implies} A(r,R) \cong \mathscr{R}_{r,R}/\big((u,0) \sim (u,2\pi)\big).$$
$$W = 2\pi \mathrm{mod}(A(r,R)) \overset{2.1.6}{\implies} \frac{W}{2\pi} = \mathrm{mod}(A(r,R)).$$

$\square$

Figure 1 records the quotient.

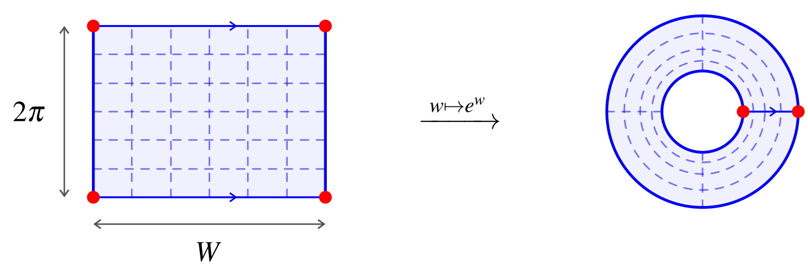

FIGURE 1. The exponential map identifies the horizontal edges, sends the vertical edges to the boundary circles, and carries the rectangular mesh to a logarithmic polar mesh.

Corollary 2.1.8 (Fundamental group, [7, 9])**.** *Every nondegenerate annulus A satisfies $\pi_1(A) \cong \mathbb{Z}$.*

Proof.

$$A \cong A(q,1) \overset{2.1.2}{\implies} \pi_1(A) \cong \pi_1(A(q,1)) \overset{2.1.6}{\implies} \mathrm{Deck}(p) \cong \mathbb{Z}.$$

$\square$

3. EXTREMAL LENGTH AND CAPACITY

The conformal modulus has two exact variational realizations: a length–area problem for transverse curve families and a Dirichlet problem for a condenser [2, 15, 20, 22].

3.1. **Dual variational principles.**

Definition 3.1.1 (Variational data)**.** Let $\Omega \subset \mathbb{C}$, let Γ be a family of rectifiable curves in Ω, and let $\rho : \Omega \to [0,\infty]$ be Borel with $0 < \mathrm{Area}_\rho(\Omega) < \infty$. Set

$$L_\rho(\Gamma) := \inf_{\gamma\in\Gamma} \int_\gamma \rho\,|dz|, \qquad \mathrm{Area}_\rho(\Omega) := \int_\Omega \rho^2\,dx\,dy, \qquad \mathrm{Ext}_\Omega(\Gamma) := \sup_\rho \frac{L_\rho(\Gamma)^2}{\mathrm{Area}_\rho(\Omega)}.$$

For a nondegenerate annulus A with boundary ends $\partial_0 A, \partial_1 A$, let $\mathscr{U}(A)$ be the Dirichlet class with traces 0 and 1 on those ends, and put

$$\mathscr{E}_A(u) := \int_A |\nabla u|^2\,dx\,dy, \qquad \mathrm{cap}(A) := \inf_{u\in\mathscr{U}(A)} \mathscr{E}_A(u).$$

The traces are read on a round model furnished by 2.1.2; interchanging the ends replaces u by $1-u$. We use the unnormalized capacity convention [15, 17, 21].

Proposition 3.1.2 (Conformal invariance)**.** *Let $\phi : \Omega \to \Omega'$ be biholomorphic. Then*

$$\mathrm{Ext}_\Omega(\Gamma) = \mathrm{Ext}_{\Omega'}\big(\phi(\Gamma)\big), \qquad \mathscr{E}_\Omega(v\circ\phi) = \mathscr{E}_{\Omega'}(v), \qquad \mathrm{cap}(\Omega) = \mathrm{cap}(\Omega')$$

whenever the last expression concerns annuli [2, 15, 22].

Proof. Pull back an arbitrary density σ and potential v by $\rho = (\sigma\circ\phi)|\phi'|$ and $u = v\circ\phi$.

$$\int_\gamma \rho\,|dz| = \int_{\phi\circ\gamma} \sigma\,|dw|, \qquad \mathrm{Area}_\rho(\Omega) = \int_\Omega (\sigma^2\circ\phi)|\phi'|^2\,dx\,dy = \mathrm{Area}_\sigma(\Omega').$$

Consequently,

$$\frac{L_\rho(\Gamma)^2}{\mathrm{Area}_\rho(\Omega)} = \frac{L_\sigma(\phi(\Gamma))^2}{\mathrm{Area}_\sigma(\Omega')} \overset{3.1.1}{\Longrightarrow} \mathrm{Ext}_\Omega(\Gamma) \ge \mathrm{Ext}_{\Omega'}\big(\phi(\Gamma)\big),$$

while

$$|\nabla(v\circ\phi)|^2 = (|\nabla v|^2\circ\phi)|\phi'|^2, \qquad J_\phi = |\phi'|^2 \qquad \Longrightarrow \qquad \mathscr{E}_\Omega(v\circ\phi) = \mathscr{E}_{\Omega'}(v).$$

Applying the same identities to ϕ^{-1}, and replacing v by $1-v$ if ϕ exchanges the ends, proves all three claims. □

Definition 3.1.3 (Dual curve families)**.** For a nondegenerate annulus A, let $\Gamma_{\mathrm{rad}}(A)$ be the rectifiable arcs joining its two boundary ends, with interiors in A, and let

$$\Gamma_{\mathrm{sep}}(A) := \big\{\gamma : S^1 \to A : \gamma \text{ is rectifiable and } [\gamma] \ne 0 \text{ in } \pi_1(A)\big\}.$$

Thus $\Gamma_{\mathrm{sep}}(A)$ is the essential, or separating, family; the identification $\pi_1(A) \cong \mathbb{Z}$ is recorded in 2.1.8 [2, 15, 20].

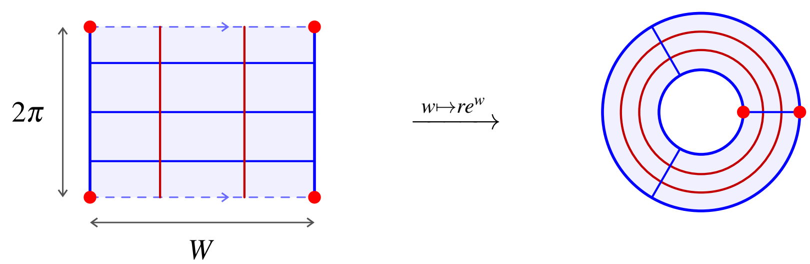

FIGURE 2. Under $w \mapsto re^w$, blue crosscuts project to radial curves and red vertical fibers to essential loops.

Theorem 3.1.4 (Variational characterization of modulus)**.** *Let A be a nondegenerate annulus and set $m := \operatorname{mod}(A)$. Then*

$$\operatorname{Ext}_A\big(\Gamma_{\mathrm{rad}}(A)\big) = m, \qquad \operatorname{Ext}_A\big(\Gamma_{\mathrm{sep}}(A)\big) = \operatorname{cap}(A) = m^{-1}.$$

In particular,

$$\operatorname{cap}(A) = \operatorname{Ext}_A\big(\Gamma_{\mathrm{sep}}(A)\big) = \operatorname{Ext}_A\big(\Gamma_{\mathrm{rad}}(A)\big)^{-1}, \qquad \operatorname{Ext}_A\big(\Gamma_{\mathrm{rad}}(A)\big)\operatorname{Ext}_A\big(\Gamma_{\mathrm{sep}}(A)\big) = 1.$$

If $A = A(r,R)$ and $W := \log(R/r)$, then $\rho_*(z) := |z|^{-1}$ is extremal for both curve families, and the unique capacity minimizer is

$$u_*(z) := \frac{\log(|z|/r)}{W}.$$

These normalizations are classical [2, 15, 17, 20, 22].

Proof. Uniformization and conformal invariance reduce all three problems to one Euclidean cylinder.

$$C_W := (0,W)\times\big(\mathbb{R}/(2\pi\mathbb{Z})\big) \overset{2.1.7}{\Longleftrightarrow} A(r,R), \qquad W = \log(R/r) = 2\pi m \overset{2.1.4}{\Longleftrightarrow} m = \frac{W}{2\pi}.$$

For an admissible density $\widetilde{\rho}$ on C_W, the horizontal segments and vertical fibers in 2 give

$$2\pi L_{\widetilde{\rho}}(\widetilde{\Gamma}_{\mathrm{rad}}) \le \int_0^{2\pi}\int_0^W \widetilde{\rho}\,dx\,dy \le (2\pi W)^{1/2}\operatorname{Area}_{\widetilde{\rho}}(C_W)^{1/2} \qquad \Longrightarrow \qquad \frac{L_{\widetilde{\rho}}(\widetilde{\Gamma}_{\mathrm{rad}})^2}{\operatorname{Area}_{\widetilde{\rho}}(C_W)} \le \frac{W}{2\pi},$$

$$W L_{\widetilde{\rho}}(\widetilde{\Gamma}_{\mathrm{sep}}) \le \int_0^W\int_0^{2\pi} \widetilde{\rho}\,dy\,dx \le (2\pi W)^{1/2}\operatorname{Area}_{\widetilde{\rho}}(C_W)^{1/2} \qquad \Longrightarrow \qquad \frac{L_{\widetilde{\rho}}(\widetilde{\Gamma}_{\mathrm{sep}})^2}{\operatorname{Area}_{\widetilde{\rho}}(C_W)} \le \frac{2\pi}{W}.$$

For $\widetilde{\rho}_* \equiv 1$, connecting curves have length at least W, while every essential loop lifts with endpoint displacement $2\pi i k$, $k \in \mathbb{Z}\setminus\{0\}$, and hence has length at least 2π; horizontal and vertical fibers attain both bounds. Therefore

$$L_{\widetilde{\rho}_*}(\widetilde{\Gamma}_{\mathrm{rad}}) = W, \quad L_{\widetilde{\rho}_*}(\widetilde{\Gamma}_{\mathrm{sep}}) = 2\pi, \quad \operatorname{Area}_{\widetilde{\rho}_*}(C_W) = 2\pi W \Longrightarrow \widetilde{\rho}_* \equiv 1 \overset{3.1.1}{\Longrightarrow} \operatorname{Ext}_{C_W}(\widetilde{\Gamma}_{\mathrm{rad}}) = \frac{W}{2\pi}, \quad \operatorname{Ext}_{C_W}(\widetilde{\Gamma}_{\mathrm{sep}}) = \frac{2\pi}{W}.$$

For $\widetilde{u} \in \mathscr{U}(C_W)$, the trace values and Cauchy–Schwarz yield, for almost every y,

$$1 = \int_0^W \partial_x\widetilde{u}(x,y)\,dx \le W^{1/2}\left(\int_0^W |\partial_x\widetilde{u}(x,y)|^2\,dx\right)^{1/2} \Longrightarrow \mathscr{E}_{C_W}(\widetilde{u}) \ge \int_0^{2\pi}\int_0^W |\partial_x\widetilde{u}|^2\,dx\,dy \ge \frac{2\pi}{W}.$$

The affine potential attains this bound, and its equality conditions give uniqueness:

$$\widetilde{u}_*(x,y) := \frac{x}{W}, \qquad \mathscr{E}_{C_W}(\widetilde{u}_*) = \int_0^{2\pi}\int_0^W \frac{1}{W^2}\,dx\,dy = \frac{2\pi}{W}.$$

$$\mathscr{E}_{C_W}(\widetilde{u}) = \frac{2\pi}{W} \qquad \Longrightarrow \qquad \partial_y\widetilde{u} = 0, \quad \partial_x\widetilde{u} = W^{-1} \qquad \Longrightarrow \qquad \widetilde{u} = \widetilde{u}_*.$$

Transporting these extremizers through the logarithmic model gives

$$\widetilde{\rho}_* \equiv 1 \overset{2.1.6}{\Longleftrightarrow} \rho_*(z) = \frac{1}{|z|}, \qquad \widetilde{u}_*(x,y) = \frac{x}{W} \overset{2.1.7}{\Longleftrightarrow} u_*(z) = \frac{\log(|z|/r)}{W} \Longrightarrow \frac{W}{2\pi} = m,$$

$$\frac{2\pi}{W} = m^{-1} \overset{2.1.2,\,3.1.2}{\Longrightarrow} \operatorname{Ext}_A(\Gamma_{\mathrm{rad}}) = m, \quad \operatorname{Ext}_A(\Gamma_{\mathrm{sep}}) = \operatorname{cap}(A) = m^{-1}.$$

This proves the theorem. $\square$

4. Coverings and Fourier analysis

Fix $A := A(r,R)$ and put

$$s := \sqrt{rR}, \qquad W := \log(R/r) \overset{2.1.4}{\iff} W = 2\pi \operatorname{mod}(A), \qquad S_W := \{w \in \mathbb{C} : |\Re w| < W/2\}.$$

By Proposition 2.1.6, $p_s(w) := se^w$ has deck generator $T(w) := w + 2\pi i$; pullback by p_s converts Laurent monomials into periodic Fourier modes [9, 7, 19].

4.1. Mode damping.

Proposition 4.1.1 (Periodic descent, [9, 7])**.** *For the centered cover $p_s : S_W \to A$, composition with p_s gives*

$$\operatorname{Hol}(A, \mathbb{C}) \xrightarrow{\cong} \big\{F \in \operatorname{Hol}(S_W, \mathbb{C}) : F \circ T = F\big\}, \qquad f \longmapsto F := f \circ p_s.$$

Equivalently, the following diagram commutes, and the dashed arrow is unique.

$$\begin{tikzcd}
S_W \arrow[r, "T"] \arrow[d, "p_s"'] \arrow[rr, bend left, "F"] & S_W \arrow[r, "F"] \arrow[d, "p_s"] & \mathbb{C} \\
A \arrow[r, equal] & A \arrow[ur, dashed, "f"'] &
\end{tikzcd}$$

Moreover,

$$F'(w) = se^w f'(se^w), \qquad z = p_s(w) \qquad \Longrightarrow \qquad zf'(z) = F'(w).$$

Proof. The fiber relation from Proposition 2.1.6 governs the descent.

$$p_s \circ T = p_s, \qquad F = f \circ p_s \qquad \Longrightarrow \qquad F \circ T = f \circ p_s \circ T = f \circ p_s = F.$$

$$p_s(w_1) = p_s(w_2) \overset{2.1.6}{\iff} w_2 - w_1 \in 2\pi i \mathbb{Z}, \qquad F \circ T = F \qquad \Longrightarrow \qquad F(w_2) = F(w_1).$$

$$f(p_s(w)) := F(w), \qquad p_s|_U : U \xrightarrow{\cong} V \qquad \Longrightarrow \qquad f|_V = F \circ (p_s|_U)^{-1} \in \operatorname{Hol}(V, \mathbb{C}).$$

$$f_1 \circ p_s = f_2 \circ p_s, \quad p_s(S_W) = A \qquad \Longrightarrow \qquad f_1 = f_2, \qquad F = f \circ p_s \implies F'(w) = se^w f'(se^w).$$

This proves the descent bijection and derivative identity. □

Theorem 4.1.2 (Fourier–Laurent correspondence, [7, 18, 19])**.** *Let $f \in \operatorname{Hol}(A, \mathbb{C})$ and let $F := f \circ p_s$. There are unique coefficients $(a_n)_{n \in \mathbb{Z}}$ and $(b_n)_{n \in \mathbb{Z}}$ such that*

$$f(z) = \sum_{n \in \mathbb{Z}} a_n z^n, \qquad F(w) = \sum_{n \in \mathbb{Z}} b_n e^{nw}, \qquad b_n = s^n a_n,$$

with absolute and locally uniform convergence. For $r < \rho < R$, $|x| < W/2$, and $n \in \mathbb{Z}$,

$$a_n = \frac{1}{2\pi i} \int_{|z|=\rho} \frac{f(z)}{z^{n+1}}\, dz, \qquad b_n = \frac{e^{-nx}}{2\pi} \int_0^{2\pi} F(x+iy) e^{-iny}\, dy.$$

Proof. Cauchy's theorem and Laurent's theorem identify the two expansions.

$$a_n(\rho) := \frac{1}{2\pi i} \int_{|z|=\rho} \frac{f(z)}{z^{n+1}}\, dz, \qquad a_n(\rho_2) - a_n(\rho_1) = \frac{1}{2\pi i} \int_{\partial A(\rho_1, \rho_2)} \frac{f(z)}{z^{n+1}}\, dz = 0.$$

$$f(z) = \sum_{n \in \mathbb{Z}} a_n z^n \quad \Longrightarrow \quad F(w) = f(se^w) = \sum_{n \in \mathbb{Z}} (s^n a_n) e^{nw} \quad \Longrightarrow \quad b_n = s^n a_n.$$

$$z = se^{x+iy}, \qquad dz = ise^{x+iy}\, dy \qquad \Longrightarrow \qquad b_n = s^n a_n = \frac{e^{-nx}}{2\pi} \int_0^{2\pi} F(x+iy) e^{-iny}\, dy.$$

$$K \Subset S_W, \quad p_s(K) \Subset A \qquad \Longrightarrow \qquad \sum_{n \in \mathbb{Z}} a_n (se^w)^n \longrightarrow f(se^w) = F(w) \quad \text{uniformly on } K.$$

This proves the Fourier–Laurent dictionary. □

Theorem 4.1.3 (Sharp mode damping, [7, 8])**.** *Let $f \in \mathrm{Hol}(A,\mathbb{C})$ satisfy $\|f\|_{L^\infty(A)} \le M$, and let b_n be the centered coefficients in Theorem 4.1.2. Every mode satisfies*

$$|b_n| \le Me^{-|n|W/2} = Me^{-\pi|n|\mathrm{mod}(A)} \qquad (n \in \mathbb{Z}),$$

and the exponent and constant are sharp for each $n \neq 0$.

Proof. The coefficient line may be moved to the favorable side of the strip.

$$|b_n| \overset{4.1.2}{\Longrightarrow} \frac{e^{-nx}}{2\pi}\left|\int_0^{2\pi} F(x+iy)e^{-iny}\,dy\right| \le \frac{e^{-nx}}{2\pi}\int_0^{2\pi} |F(x+iy)|\,dy \le Me^{-nx}.$$

$$n \ge 0, \quad x \uparrow W/2 \quad \Longrightarrow \quad |b_n| \le Me^{-nW/2}, \qquad n < 0, \quad x \downarrow -W/2 \quad \Longrightarrow \quad |b_n| \le Me^{-|n|W/2}.$$

$$W = 2\pi\mathrm{mod}(A) \overset{2.1.4}{\Longrightarrow} e^{-|n|W/2} = e^{-\pi|n|\mathrm{mod}(A)}, \qquad \frac{s}{R} = \frac{r}{s} = e^{-W/2}.$$

$$f_k^+(z) := M\left(\frac{z}{R}\right)^k, \qquad f_k^-(z) := M\left(\frac{r}{z}\right)^k, \qquad \|f_k^\pm\|_{L^\infty(A)} = M, \qquad b_k^+ = b_{-k}^- = Me^{-kW/2} \quad (k \ge 1).$$

This proves the sharp damping estimate. □

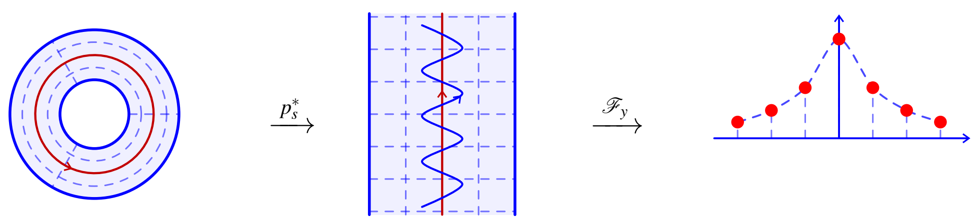

FIGURE 3. Pullback by $p_s(w) = se^w$ sends a_nz^n to b_ne^{nw}, where $b_n = s^na_n$, and the centered strip yields $|b_n| \le Me^{-|n|W/2}$.

Corollary 4.1.4 (Core-circle rigidity)**.** *Let $m := \mathrm{mod}(A)$, $\delta := e^{-\pi m}$, and $\|f\|_{L^\infty(A)} \le M$. On the core circle $|z| = s$ in Figure 3,*

$$\sup_{|z|=s} |f(z) - a_0| \le \frac{2M\delta}{1-\delta}, \qquad \sup_{|z|=s} |zf'(z)| \le \frac{2M\delta}{(1-\delta)^2}.$$

For $A = A(q,1)$, Proposition 2.1.4 gives

$$s = \sqrt{q}, \qquad \delta = \sqrt{q}, \qquad \sup_{|z|=\sqrt{q}} |f(z) - a_0| \le \frac{2M\sqrt{q}}{1-\sqrt{q}}, \qquad \sup_{|z|=\sqrt{q}} |zf'(z)| \le \frac{2M\sqrt{q}}{(1-\sqrt{q})^2}.$$

Proof. The centered Fourier expansion and Theorem 4.1.3 control both tails.

$$z = se^{iy} \qquad \Longrightarrow \qquad f(z) - a_0 = \sum_{n\neq 0} b_ne^{iny}, \qquad zf'(z) = \sum_{n\neq 0} nb_ne^{iny}.$$

$$\sup_{|z|=s} |f(z) - a_0| \le \sum_{n\neq 0} |b_n| \overset{4.1.3}{\Longrightarrow} M\sum_{n\neq 0} \delta^{|n|} = 2M\sum_{n\ge 1} \delta^n = \frac{2M\delta}{1-\delta}.$$

$$\sup_{|z|=s} |zf'(z)| \le \sum_{n\neq 0} |n|\,|b_n| \overset{4.1.3}{\Longrightarrow} 2M\sum_{n\ge 1} n\delta^n = \frac{2M\delta}{(1-\delta)^2}.$$

This proves exponential rigidity on the core circle. □

Thus the cylinder length $W = 2\pi\mathrm{mod}(A)$ is the exact spectral separation scale, and every nonconstant core mode loses the factor $e^{-\pi|n|\mathrm{mod}(A)}$ [7, 8].

5. Hyperbolic geometry and Schwarz–Pick

The centered cover from Proposition 2.1.6 turns the complete curvature--1 metric into an explicit transverse density. Its core geometry and Schwarz–Pick contraction make the dependence on $\mathrm{mod}(A)$ intrinsic [6, 5, 12].

5.1. **Intrinsic estimates.**

Definition 5.1.1 (Hyperbolic metric, [6, 5])**.** Let $\Omega \subset \mathbb{C}$ have universal cover $\mathbb{D}$, and let $p : \mathbb{D} \to \Omega$ be a covering. Its Poincaré density, curve length, and distance are normalized by

$$\rho_{\mathbb{D}}(\zeta) := \frac{2}{1-|\zeta|^2}, \qquad \rho_{\mathbb{H}}(\zeta) := \frac{1}{\Im(\zeta)}, \qquad \rho_\Omega(p(\zeta))|p'(\zeta)| = \rho_{\mathbb{D}}(\zeta), \qquad ds_\Omega := \rho_\Omega(z)|dz|.$$

$$\ell_\Omega(\gamma) := \int_\gamma \rho_\Omega(z)|dz|, \qquad d_\Omega(z_0,z_1) := \inf_{\gamma: z_0 \rightsquigarrow z_1} \ell_\Omega(\gamma).$$

Theorem 5.1.2 (Explicit annular density, [6, 5, 12])**.** *Let $A = A(r,R)$, $s := \sqrt{rR}$, $W := \log(R/r)$, and $m := \mathrm{mod}(A)$. Then*

$$W = 2\pi m, \qquad \rho_A(z) = \frac{\pi}{W|z|}\sec\left(\frac{\pi}{W}\log\frac{|z|}{s}\right) = \frac{\pi}{W|z|}\csc\left(\frac{\pi}{W}\log\frac{|z|}{r}\right), \qquad \rho_A(se^{i\theta}) = \frac{1}{2ms}.$$

Proof. We pull back the half-plane metric through the centered cover.

$$S_W := \{w \in \mathbb{C} : |\Re(w)| < W/2\}, \qquad \Phi(w) := i\exp(\pi i w/W), \qquad \Phi : S_W \xrightarrow{\sim} \mathbb{H}.$$

$$x := \Re(w), \qquad \Im(\Phi(w)) = |\Phi(w)|\cos(\pi x/W), \qquad |\Phi'(w)| = \frac{\pi}{W}|\Phi(w)| \implies \rho_{S_W}(w) = \frac{\pi}{W}\sec(\pi x/W).$$

$$p_s(w) := se^w, \qquad |p_s'(w)| = |p_s(w)| \overset{2.1.6,5.1.1}{\Longrightarrow} \rho_A(p_s(w))|p_s(w)| = \rho_{S_W}(w).$$

$$x = \log\frac{|z|}{s}, \qquad \log\frac{|z|}{r} = x + \frac{W}{2}, \qquad \rho_A(z) = \frac{\pi}{W|z|}\sec\left(\frac{\pi x}{W}\right) = \frac{\pi}{W|z|}\csc\left(\frac{\pi}{W}\log\frac{|z|}{r}\right).$$

$$m = \mathrm{mod}(A) \overset{2.1.4}{\iff} W = 2\pi m, \qquad |z| = s \implies \rho_A(se^{i\theta}) = \frac{\pi}{Ws} = \frac{1}{2ms}.$$

This proves the density formula and its core value. $\square$

Proposition 5.1.3 (Intrinsic geometry, [11, 5, 12])**.** *Let $A = A(r,R)$, $s := \sqrt{rR}$, $m := \mathrm{mod}(A)$, $W := \log(R/r) = 2\pi m$, and*

$$\Delta_A(z) := \min\left\{\log\frac{|z|}{r}, \log\frac{R}{|z|}\right\} = \frac{W}{2} - \left|\log\frac{|z|}{s}\right|.$$

The density satisfies

$$\frac{1}{|z|\Delta_A(z)} \le \rho_A(z) \le \frac{\pi}{2|z|\Delta_A(z)}.$$

For $r < \rho_1, \rho_2 < R$, $t_j := \log(\rho_j/r)$, and $\theta \in \mathbb{R}$,

$$d_A(\rho_1 e^{i\theta}, \rho_2 e^{i\theta}) = \left|\log\frac{\tan(\pi t_2/(2W))}{\tan(\pi t_1/(2W))}\right|.$$

Let $\Gamma_1(A)$ be the rectifiable closed curves representing a generator of $\pi_1(A)$. Then

$$C_* := \{|z| = s\} \text{ is a hyperbolic geodesic}, \qquad \ell_A(C_*) = \frac{2\pi^2}{W} = \frac{\pi}{m}.$$

$$\min_{\gamma \in \Gamma_1(A)} \ell_A(\gamma) = \ell_A(C_*), \qquad \ell_A(\gamma) = \ell_A(C_*) \iff \gamma \text{ parametrizes } C_* \text{ once without backtracking}.$$

Proof. Write every generator lift as $\widetilde{\gamma} = X + iY$ and use the models in Figure 4.

$$u := \frac{\pi\Delta_A(z)}{W} \in (0,\pi/2] \overset{5.1.2}{\Longrightarrow} \rho_A(z)|z|\Delta_A(z) = \frac{u}{\sin u}, \qquad \frac{2u}{\pi} \le \sin u \le u \implies 1 \le \frac{u}{\sin u} \le \frac{\pi}{2}.$$

$$\ell_A(\gamma) \ge \left|\int_{t_1}^{t_2} \frac{\pi}{W}\csc(\pi t/W)\,dt\right| = \left|\log\frac{\tan(\pi t_2/(2W))}{\tan(\pi t_1/(2W))}\right|, \qquad \gamma(\tau) = e^{(1-\tau)\log\rho_1 + \tau\log\rho_2 + i\theta} \implies \text{equality}.$$

$$\Phi(i\mathbb{R}) = i\mathbb{R}_{>0} \text{ is geodesic in } \mathbb{H} \overset{5.1.2}{\Longrightarrow} C_* = p_s(i\mathbb{R}) \text{ is geodesic}, \qquad \ell_A(C_*) = 2\pi\cdot\frac{\pi}{W} = \frac{2\pi^2}{W} = \frac{\pi}{m}.$$

$$Y(1) - Y(0) = \pm 2\pi, \qquad \rho_{S_W}(X+iY) \ge \frac{\pi}{W}, \qquad |\widetilde{\gamma}'| \ge |Y'| \implies \ell_A(\gamma) \ge \frac{\pi}{W}\int_0^1 |Y'(t)|dt \ge \frac{2\pi^2}{W}.$$

$$\ell_A(\gamma) = \frac{2\pi^2}{W} \iff X \equiv 0 \wedge Y \text{ is monotone} \iff \gamma \text{ parametrizes } C_* \text{ once without backtracking}.$$

This proves the comparison, distance, and core-minimality claims. ☐

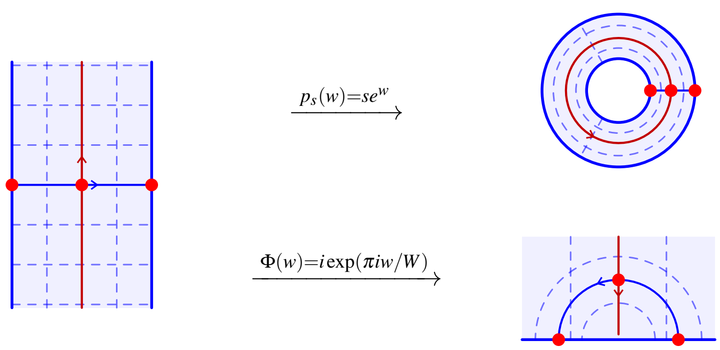

FIGURE 4. The centered strip sends its red midline to the annular core under p_s and to $i\mathbb{R}_{>0}$ under Φ; its blue transverse geodesic becomes a radius and a semicircle.

Theorem 5.1.4 (Schwarz–Pick, [1, 7, 5, 6])**.** *Let $\Omega, \Omega' \subset \mathbb{C}$ be hyperbolic domains, and let $f \in \mathrm{Hol}(\Omega,\Omega')$. Then*

$$\rho_{\Omega'}(f(z))|f'(z)| \le \rho_\Omega(z), \qquad d_{\Omega'}(f(z_0), f(z_1)) \le d_\Omega(z_0,z_1).$$

Proof. Fix $z \in \Omega$ and lift f through normalized disk covers.

$$\begin{tikzcd} (\mathbb{D},0) \arrow[r, "\widetilde{f}"] \arrow[d, "p"'] & (\mathbb{D},0) \arrow[d, "q"] \\ (\Omega,z) \arrow[r, "f"'] & (\Omega', f(z)) \end{tikzcd} \qquad q\circ\widetilde{f} = f\circ p, \qquad \widetilde{f}(0) = 0.$$

$$|\widetilde{f}'(0)| \le 1, \qquad q'(0)\widetilde{f}'(0) = f'(z)p'(0) \implies |f'(z)|\,|p'(0)| \le |q'(0)|.$$

$$p,q \text{ universal covers} \overset{5.1.1}{\Longrightarrow} \rho_\Omega(z)|p'(0)| = \rho_{\mathbb{D}}(0) = 2 = \rho_{\Omega'}(f(z))|q'(0)| \implies \rho_{\Omega'}(f(z))|f'(z)| \le \rho_\Omega(z).$$

$$\ell_{\Omega'}(f\circ\gamma) = \int_\gamma \rho_{\Omega'}(f(z))|f'(z)|\,|dz| \le \int_\gamma \rho_\Omega(z)|dz| = \ell_\Omega(\gamma) \implies d_{\Omega'}(f(z_0),f(z_1)) \le d_\Omega(z_0,z_1).$$

This proves the infinitesimal and global contractions. ☐

Corollary 5.1.5 (Annulus-to-disk estimates)**.** *Let $A = A(r,R)$, $s := \sqrt{rR}$, $m := \operatorname{mod}(A)$, $W := \log(R/r) = 2\pi m$, and $f \in \operatorname{Hol}(A,\mathbb{D})$. For*

$$\vartheta(\theta,\varphi) := \min_{k\in\mathbb{Z}} |\theta - \varphi + 2\pi k|,$$

one has

$$\frac{|zf'(z)|}{1-|f(z)|^2} \le \frac{\pi}{2W}\sec\left(\frac{\pi}{W}\log\frac{|z|}{s}\right), \qquad |f'(z)| \le \frac{\pi(1-|f(z)|^2)}{4|z|\Delta_A(z)} \le \frac{\pi}{4|z|\Delta_A(z)}.$$

$$\frac{|se^{i\theta} f'(se^{i\theta})|}{1-|f(se^{i\theta})|^2} \le \frac{1}{4m}, \qquad d_{\mathbb{D}}(f(se^{i\theta}), f(se^{i\varphi})) \le \frac{\vartheta(\theta,\varphi)}{2m}.$$

Proof. We insert the explicit density into Schwarz–Pick.

$$f \in \operatorname{Hol}(A,\mathbb{D}) \overset{5.1.4}{\Longrightarrow} \frac{2|f'(z)|}{1-|f(z)|^2} \le \rho_A(z) \overset{5.1.2}{\Longrightarrow} \frac{|zf'(z)|}{1-|f(z)|^2} \le \frac{\pi}{2W}\sec\left(\frac{\pi}{W}\log\frac{|z|}{s}\right).$$

$$\frac{2|f'(z)|}{1-|f(z)|^2} \le \rho_A(z) \overset{5.1.3}{\Longrightarrow} \frac{2|f'(z)|}{1-|f(z)|^2} \le \frac{\pi}{2|z|\Delta_A(z)} \implies |f'(z)| \le \frac{\pi(1-|f(z)|^2)}{4|z|\Delta_A(z)}.$$

$$z = se^{i\theta}, \qquad W = 2\pi m \implies \frac{|zf'(z)|}{1-|f(z)|^2} \le \frac{\pi}{2W} = \frac{1}{4m}.$$

$$\gamma_{\theta,\varphi} \subset C_* \text{ is the shorter arc}, \qquad \ell_A(\gamma_{\theta,\varphi}) = \frac{\pi}{W}\vartheta(\theta,\varphi) = \frac{\vartheta(\theta,\varphi)}{2m}.$$

$$f \in \operatorname{Hol}(A,\mathbb{D}) \overset{5.1.4}{\Longrightarrow} d_{\mathbb{D}}(f(se^{i\theta}), f(se^{i\varphi})) \le d_A(se^{i\theta}, se^{i\varphi}) \le \ell_A(\gamma_{\theta,\varphi}) = \frac{\vartheta(\theta,\varphi)}{2m}.$$

This proves the derivative and angular estimates. □

Together with Theorem 4.1.3, these estimates separate the intrinsic scale m^{-1} from the Fourier scale $e^{-\pi m}$ [6, 12].

6. Holomorphic maps between annuli

Throughout, $A := A(r,R)$, $A' := A(r',R')$, $s := \sqrt{rR}$, $s' := \sqrt{r'R'}$, $W := \log(R/r)$, and $W' := \log(R'/r')$, with centered covers $p_s : S_W \to A$, $p_{s'} : S_{W'} \to A'$, and deck generator $T(w) := w + 2\pi i$; properness forces exact modulus scaling [3, 7, 9].

6.1. **Degree and rigidity.**

Definition 6.1.1 (Properness and degree, [7, 9])**.** For a continuous map $f : A \to A'$, properness and degree under the counterclockwise identifications in Corollary 2.1.8 are specified by

$$f \text{ proper} \iff f^{-1}(K) \Subset A \quad \text{for every } K \Subset A', \qquad f_* : \mathbb{Z} \longrightarrow \mathbb{Z}, \qquad k \longmapsto \deg(f)\cdot k.$$

For $f \in \operatorname{Hol}(A,A')$ and a piecewise C^1 counterclockwise generator γ,

$$\deg(f) = \operatorname{wind}_0(f\circ\gamma) = \frac{1}{2\pi i}\int_\gamma \frac{f'(z)}{f(z)}\,dz.$$

Proposition 6.1.2 (Equivariant lift, [7, 9])**.** *Every $f \in \operatorname{Hol}(A,A')$ admits a holomorphic lift $F : S_W \to S_{W'}$ and a unique $n \in \mathbb{Z}$ such that*

$$\begin{tikzcd} S_W \arrow[r, "F"] \arrow[d, "p_s"'] & S_{W'} \arrow[d, "p_{s'}"] \\ A \arrow[r, "f"'] & A' \end{tikzcd} \qquad F\circ T = T^n \circ F, \qquad n = \deg(f).$$

Moreover,

$$|\deg(f)|\cdot\operatorname{mod}(A)\le\operatorname{mod}(A'),\qquad \deg(f)=0 \iff f=e^g \quad \textit{for some } g\in\operatorname{Hol}(A,\mathbb{C}).$$

Proof. Lift $f\circ p_s$ through the logarithmic cover of A' and average its real part across one deck period.

$$p_{s'}\circ F=f\circ p_s,\qquad p_s\circ T=p_s \implies F(T(w))-F(w)\in 2\pi i\mathbb{Z}.$$

$$\frac{F(T(w))-F(w)}{2\pi i}\in\operatorname{Hol}(S_W,\mathbb{Z}),\qquad S_W \text{ connected} \implies F\circ T=T^n\circ F \overset{6.1.1}{\iff} n=\deg(f).$$

$$M(x):=\frac{1}{2\pi}\int_0^{2\pi}\Re(F(x+iy))\,dy,\qquad M'(x)=\frac{\Im(F(x+2\pi i))-\Im(F(x))}{2\pi}=n,$$

$$|\Re(F(w))|<\frac{W'}{2},\qquad M(x)=n\cdot x+a \implies |n|\cdot W\le W' \overset{2.1.4}{\iff} |n|\cdot\operatorname{mod}(A)\le\operatorname{mod}(A').$$

$$n=0 \iff F\circ T=F \overset{4.1.1}{\iff} F=G\circ p_s \iff f=e^g,\qquad g:=\log(s')+G.$$

This proves the lift, degree, modulus, and logarithm claims. □

Theorem 6.1.3 (Proper-map classification, [3, 7, 9])**.** *Let $f\in\operatorname{Hol}(A,A')$ be proper and put $n:=\deg(f)$. Then*

$$n\neq 0,\qquad f(z)=c\cdot z^n\quad (c\in\mathbb{C}\setminus\{0\}),\qquad \operatorname{mod}(A')=|n|\cdot\operatorname{mod}(A),$$
$$\{|c|r^n,|c|R^n\}=\{r',R'\},\qquad f:A\to A' \textit{ is an } |n|\textit{-sheeted unbranched covering}.$$

Proof. Let F be the lift from Proposition 6.1.2; properness excludes constants, while openness, closedness, and connected collars outside compact middle preimages give the uniform end limits below.

$$\Re(F(x+iy))\longrightarrow \varepsilon_\pm\frac{W'}{2}\quad (x\to\pm W/2),\qquad \varepsilon_+=-\varepsilon_-=:\varepsilon\in\{1,-1\}.$$

$$h(z):=\log\frac{|f(z)|}{s'}-\varepsilon\frac{W'}{W}\log\frac{|z|}{s},\qquad \Delta h=0,\qquad h(z)\to 0\quad (z\to\partial A) \implies h=0.$$

$$h=0 \implies \Re(F(w))=\varepsilon\frac{W'}{W}\Re(w),\qquad \Re\left(F(w)-\varepsilon\frac{W'}{W}w\right)=0 \implies F(w)=\varepsilon\frac{W'}{W}w+i\theta\quad(\theta\in\mathbb{R}).$$

$$F\circ T=T^n\circ F \overset{6.1.2}{\implies} 2\pi i n=F(T(w))-F(w)=2\pi i\varepsilon\frac{W'}{W} \implies n=\varepsilon\frac{W'}{W}\neq 0,\qquad W'=|n|\cdot W.$$

$$z=p_s(w) \implies f(z)=p_{s'}(F(w))=s'e^{i\theta}\left(\frac{z}{s}\right)^n=c\cdot z^n,\qquad c:=s'e^{i\theta}s^{-n}\neq 0,\qquad f'(z)=n\cdot c\cdot z^{n-1}\neq 0.$$

This proves the classification, covering, and modulus law. □

Corollary 6.1.4 (Existence and self-maps)**.** *Let X,Y be nondegenerate annuli and $n\in\mathbb{Z}\setminus\{0\}$. Then*

$$\exists\, f\in\operatorname{Hol}(X,Y) \textit{ proper with } \deg(f)=n \iff \operatorname{mod}(Y)=|n|\cdot\operatorname{mod}(X).$$

Every proper holomorphic self-map of X is biholomorphic; for a round annulus,

$$f:A(r,R)\to A(r,R) \textit{ proper} \iff f(z)=e^{i\theta}z \quad\textit{or}\quad f(z)=e^{i\theta}\frac{rR}{z}\quad(\theta\in\mathbb{R}).$$

Proof. Choose degree-one uniformizations $\phi:X\to A(q,1)$ and $\psi:Y\to A(q',1)$ by Theorem 2.1.2, postcomposing with $z\mapsto q/z$ and $z\mapsto q'/z$ if necessary.

$$\widetilde{f}:=\psi\circ f\circ\phi^{-1},\qquad \deg(\widetilde{f})=n,\qquad f \text{ proper} \iff \widetilde{f} \text{ proper} \overset{6.1.3}{\implies} \operatorname{mod}(Y)=|n|\cdot\operatorname{mod}(X).$$

$$\operatorname{mod}(Y)=|n|\cdot\operatorname{mod}(X) \overset{2.1.4}{\iff} q'=q^{|n|}.$$

$$\mu_n(z):=\begin{cases} z^n, & n>0,\\ q^{|n|}\cdot z^n, & n<0,\end{cases}\qquad \mu_n:A(q,1)\to A(q',1) \text{ proper},\qquad \deg(\mu_n)=n.$$

$$f_n:=\psi^{-1}\circ\mu_n\circ\phi:X\to Y,\qquad f_n \text{ proper},\qquad \deg(f_n)=n.$$

$$X=Y \implies |n|=1 \overset{6.1.3}{\implies} f\in \mathrm{Aut}(X), \qquad X=A(r,R) \implies f(z)=e^{i\theta}z \quad \text{or} \quad f(z)=e^{i\theta}\frac{rR}{z}.$$

This proves existence and self-map rigidity. □

7. Boundary interaction and degeneration

For $A:=A(r,R)$, put $m:=\mathrm{mod}(A)$, $W:=\log(R/r)=2\pi m$, and $t:=\log(|z|/r)$; constant boundary data propagates linearly in t/W, whereas nonconstant modes decay exponentially with their distance from the supplying boundary [2, 17].

7.1. **Transfer and limiting regimes.**

Theorem 7.1.1 (Annular boundary transfer, [2, 17, 18])**.** *Let $\varphi_r,\varphi_R\in C(\mathbb{R}/(2\pi\mathbb{Z}),\mathbb{R})$. Define their Fourier coefficients and, for $n\neq 0$, the transfer factors by*

$$\widehat{\varphi}_j(n):=\frac{1}{2\pi}\int_0^{2\pi}\varphi_j(\theta)e^{-in\theta}\,d\theta, \qquad P_n^r(t):=\frac{\sinh(|n|(W-t))}{\sinh(|n|W)}, \qquad P_n^R(t):=\frac{\sinh(|n|t)}{\sinh(|n|W)}.$$

The unique harmonic $u\in C^2(A)\cap C(\overline{A})$ with boundary values φ_r,φ_R satisfies, for $0<t<W$,

$$u(re^{t+i\theta})=\left(1-\frac{t}{W}\right)\widehat{\varphi}_r(0)+\frac{t}{W}\widehat{\varphi}_R(0)+\sum_{n\neq 0}\left(P_n^r(t)\widehat{\varphi}_r(n)+P_n^R(t)\widehat{\varphi}_R(n)\right)e^{in\theta},$$

where the series converges absolutely and locally uniformly. The boundary-component harmonic measures and transfer bounds are

$$\omega_R(re^{t+i\theta})=\frac{t}{W}, \qquad \omega_r(re^{t+i\theta})=1-\frac{t}{W}, \qquad \omega_R=u_* \overset{3.1.4}{\implies} \mathscr{E}_A(\omega_R)=\mathrm{cap}(A)=m^{-1}.$$

$$0\le P_n^r(t)\le e^{-|n|t}, \qquad 0\le P_n^R(t)\le e^{-|n|(W-t)}, \qquad P_n^r(W/2)=P_n^R(W/2)=\frac{1}{2\cosh(|n|W/2)}\le e^{-\pi|n|m}.$$

Proof. It suffices to solve trigonometric-polynomial data and use Fejér approximation.

$$v(t,\theta):=u(re^{t+i\theta}), \qquad \Delta u=0 \iff \left(\frac{\partial^2}{\partial t^2}+\frac{\partial^2}{\partial\theta^2}\right)v=0, \qquad v(t,\theta)=\sum_{n\in\mathbb{Z}}v_n(t)e^{in\theta}.$$

$$v_0''=0 \implies v_0(t)=\left(1-\frac{t}{W}\right)\widehat{\varphi}_r(0)+\frac{t}{W}\widehat{\varphi}_R(0), \qquad v_n''-n^2v_n=0 \implies v_n(t)=P_n^r(t)\widehat{\varphi}_r(n)+P_n^R(t)\widehat{\varphi}_R(n).$$

$$P_n^r(t)=e^{-|n|t}\frac{1-e^{-2|n|(W-t)}}{1-e^{-2|n|W}}\le e^{-|n|t}, \qquad P_n^R(t)=e^{-|n|(W-t)}\frac{1-e^{-2|n|t}}{1-e^{-2|n|W}}\le e^{-|n|(W-t)}.$$

$$|\widehat{\varphi}_j(n)|\le\|\varphi_j\|_\infty, \qquad \Phi_N:=(\sigma_N\varphi_r,\sigma_N\varphi_R), \qquad \Phi:=(\varphi_r,\varphi_R), \qquad \|\Phi_N-\Phi\|_\infty\longrightarrow 0.$$

$$\Delta u_N=0, \qquad u_N|_{\partial A}=\Phi_N, \qquad \|u_N-u_M\|_\infty\le\|\Phi_N-\Phi_M\|_\infty \implies u_N\longrightarrow u \text{ uniformly}.$$

This proves the Dirichlet formula and its exact transfer bounds. □

Proposition 7.1.2 (Two-boundary majorization)**.** *Let $f\in\mathrm{Hol}(A,\mathbb{C})\cap C(\overline{A})$, $M_j:=\max_{|z|=j}|f(z)|>0$, and $\rho=re^t$. Then*

$$|f(\rho e^{i\theta})|\le M_r^{1-t/W}M_R^{t/W}.$$

If $f,g\in\mathrm{Hol}(A,\mathbb{C})\cap C(\overline{A})$ are nonvanishing on $\overline{A}$ and

$$\varepsilon_j:=\max_{|z|=j}\left|\log|f(z)|-\log|g(z)|\right| \qquad (j\in\{r,R\}),$$

then

$$\left|\log\left|\frac{f}{g}\right|(re^{t+i\theta})\right|\le\left(1-\frac{t}{W}\right)\varepsilon_r+\frac{t}{W}\varepsilon_R.$$

Proof. The harmonic measures in Theorem 7.1.1 are the sharp barriers.

$$\omega_r = 1-\frac{t}{W},\qquad \omega_R=\frac{t}{W},\qquad L_f := \omega_r\log M_r+\omega_R\log M_R.$$

$$\log|f|-L_f \text{ is subharmonic},\qquad \log|f|-L_f\le 0 \text{ on } \partial A \implies \log|f|\le L_f.$$

$$h:=\log\left|\frac{f}{g}\right|,\qquad H:=\omega_r\varepsilon_r+\omega_R\varepsilon_R,\qquad \Delta h=\Delta H=0,\qquad -H\le h\le H \text{ on } \partial A \implies |h|\le H.$$

This proves both weighted maximum-principle estimates. □

Theorem 7.1.3 (Long-cylinder degeneration and removability, [7, 18])**.** *Let $m_k\to\infty$, $q_k := e^{-2\pi m_k}$, $s_k:=\sqrt{q_k}$, and $f_k\in \mathrm{Hol}(A(q_k,1),\mathbb{D})$, with*

$$f_k(z)=\sum_{n\in\mathbb{Z}} a_n^{(k)}z^n,\qquad b_n^{(k)} := s_k^n a_n^{(k)}.$$

Then

$$|b_n^{(k)}|\le e^{-\pi|n|m_k},\qquad |a_n^{(k)}|\le 1\quad (n\ge 0),\qquad |a_{-n}^{(k)}|\le q_k^n\quad (n\ge 1),$$

$$\sup_{|z|=s_k}|f_k(z)-a_0^{(k)}|\le \frac{2s_k}{1-s_k}.$$

Every such sequence has a subsequence and a function $f_\infty\in\mathrm{Hol}(\mathbb{D},\mathbb{C})$, $f_\infty(\mathbb{D})\subset\overline{\mathbb{D}}$, for which

$$f_{k_j}\longrightarrow f_\infty \text{ locally uniformly on } \mathbb{D}^*,\qquad a_0^{(k_j)}\longrightarrow f_\infty(0),\qquad \sup_{\theta\in\mathbb{R}}|f_{k_j}(s_{k_j}e^{i\theta})-f_\infty(0)|\longrightarrow 0.$$

Moreover, every bounded $f(z)=\sum_{n\in\mathbb{Z}}a_nz^n\in\mathrm{Hol}(\mathbb{D}^*,\mathbb{C})$ satisfies

$$|a_{-n}|\le Mq^n = Me^{-2\pi n\,\mathrm{mod}(A(q,1))}\quad (0<q<1,\ n\ge 1),\qquad M:=\|f\|_{L^\infty(\mathbb{D}^*)},\qquad a_{-n}=0,$$

and therefore extends holomorphically across 0.

Proof. Fix $0<\rho<1$; centered damping, Montel compactness, and Cauchy's formula control both limits.

$$|b_n^{(k)}| \overset{4.1.3}{\Longrightarrow} s_k^{|n|}=e^{-\pi|n|m_k},\qquad b_n^{(k)}=s_k^n a_n^{(k)} \implies |a_n^{(k)}|\le 1\ (n\ge 0),\quad |a_{-n}^{(k)}|\le s_k^{2n}=q_k^n\ (n\ge 1).$$

$$\sup_{|z|=s_k}|f_k-a_0^{(k)}| \overset{4.1.4}{\Longrightarrow} \frac{2s_k}{1-s_k}.$$

$$a_{-n}=\frac{1}{2\pi i}\int_{|z|=q} f(z)z^{n-1}\,dz,\qquad |a_{-n}|\le Mq^n\quad (0<q<1),\qquad q\downarrow 0 \implies a_{-n}=0\quad (n\ge 1).$$

$$0<\alpha<1,\qquad A(\alpha,1)\subset A(q_k,1)\quad (k\gg 1),\qquad |f_k|\le 1 \xRightarrow{\text{Montel/diagonal}} f_{k_j}\xrightarrow{\text{loc.}} f_\infty \text{ on } \mathbb{D}^*,\qquad |f_\infty|\le 1.$$

$$c_j:=a_0^{(k_j)}=\frac{1}{2\pi i}\int_{|z|=\rho}\frac{f_{k_j}(z)}{z}\,dz\longrightarrow f_\infty(0),\qquad \sup_\theta|f_{k_j}(s_{k_j}e^{i\theta})-f_\infty(0)|\le\frac{2s_{k_j}}{1-s_{k_j}}+|c_j-f_\infty(0)|\longrightarrow 0.$$

This proves compactness, core collapse, and removability. □

For the opposite degeneration $m\downarrow 0$, let $s:=\sqrt{rR}$ and $F\in\mathrm{Hol}(A,\mathbb{D})$. The intrinsic formulas of § 5 give

$$s\rho_A(se^{i\theta})=\frac{1}{2m},\qquad \ell_A(\{|z|=s\})=\frac{\pi}{m},\qquad \frac{s|F'(se^{i\theta})|}{1-|F(se^{i\theta})|^2}\le\frac{1}{4m}.$$

The derivative order is attained up to constants by

$$0<m<\pi^{-1},\qquad N_m:=\left\lfloor\frac{1}{\pi m}\right\rfloor,\qquad G_m(z):=e^{-\pi m N_m}z^{N_m}\in\mathrm{Hol}(A(e^{-\pi m},e^{\pi m}),\mathbb{D}).$$

$$m|G_m'(1)|=mN_me^{-\pi m N_m}\longrightarrow\frac{1}{\pi e}.$$

Thus long cylinders suppress nonconstant modes at the scale $e^{-\pi m}$, whereas short cylinders have intrinsic scale m^{-1}.

8. QUASICONFORMAL STABILITY AND OUTLOOK

Quasiconformal deformation preserves annular invariants up to maximal dilatation, and $\frac{1}{2}\log \operatorname{mod}(A)$ is the exact Teichmüller coordinate [2, 13, 10].

8.1. **Sharp modulus distortion.**

Definition 8.1.1 (Quasiconformal map, [13, 4])**.** Let $\Omega, \Omega' \subset \mathbb{C}$ and $K \geq 1$. An orientation-preserving homeomorphism $f : \Omega \to \Omega'$ with $f \in W^{1,2}_{\text{loc}}(\Omega, \mathbb{C})$ is K-quasiconformal if

$$\left|\left(\frac{\partial}{\partial \bar{z}} f\right)(z)\right| \leq \frac{K-1}{K+1}\left|\left(\frac{\partial}{\partial z} f\right)(z)\right| \qquad \text{for almost every } z \in \Omega.$$

Equivalently, with $\|Df(z)\|$ the operator norm and $J_f(z) := \det(Df(z))$,

$$\|Df(z)\|^2 \leq K J_f(z) \quad \text{a.e.}, \qquad K(f) := \inf\{K \geq 1 : f \text{ is } K\text{-quasiconformal}\}.$$

Theorem 8.1.2 (Extremal-length distortion, [2, 13])**.** *Let $f : \Omega \to \Omega'$ be K-quasiconformal and let Γ be a curve family in Ω. Then*

$$\frac{1}{K}\operatorname{Ext}_\Omega(\Gamma) \leq \operatorname{Ext}_{\Omega'}(f(\Gamma)) \leq K\operatorname{Ext}_\Omega(\Gamma).$$

Proof. Fix $\rho' \geq 0$ with $0 < \operatorname{Area}_{\rho'}(\Omega') < \infty$; the ACL chain rule and a Fuglede correction control a zero-2-modulus exceptional family.

$$\ell_{\rho'}(f \circ \gamma) \leq \int_\gamma \rho'(f(z))\|Df(z)\|\,|dz| \quad (\gamma \notin \Gamma_0), \qquad h \in L^2(\Omega), \quad h \geq 0, \quad \int_\gamma h\,|dz| = \infty \quad (\gamma \in \Gamma_0).$$

$$\rho_\varepsilon(z) := \rho'(f(z))\|Df(z)\| + \varepsilon h(z), \qquad L_{\rho_\varepsilon}(\Gamma) \geq L_{\rho'}(f(\Gamma)).$$

$$\operatorname{Area}_{\rho_\varepsilon}(\Omega) \longrightarrow \int_\Omega (\rho' \circ f)^2 \|Df\|^2\, dA \leq K \int_\Omega (\rho' \circ f)^2 J_f\, dA = K\operatorname{Area}_{\rho'}(\Omega').$$

$$\operatorname{Ext}_\Omega(\Gamma) \geq \lim_{\varepsilon \downarrow 0} \frac{L_{\rho_\varepsilon}(\Gamma)^2}{\operatorname{Area}_{\rho_\varepsilon}(\Omega)} \geq \frac{1}{K}\frac{L_{\rho'}(f(\Gamma))^2}{\operatorname{Area}_{\rho'}(\Omega')} \implies \operatorname{Ext}_{\Omega'}(f(\Gamma)) \leq K\operatorname{Ext}_\Omega(\Gamma).$$

$$K(f^{-1}) = K(f) \leq K \implies \operatorname{Ext}_\Omega(\Gamma) \leq K\operatorname{Ext}_{\Omega'}(f(\Gamma)) \implies \operatorname{Ext}_{\Omega'}(f(\Gamma)) \geq K^{-1}\operatorname{Ext}_\Omega(\Gamma).$$

This proves both distortion inequalities. □

Corollary 8.1.3 (Sharp annular distortion)**.** *Let $f : A \to A'$ be K-quasiconformal, $m := \operatorname{mod}(A)$, and $m' := \operatorname{mod}(A')$. Then*

$$\frac{m}{K} \leq m' \leq Km, \qquad \frac{\operatorname{cap}(A)}{K} \leq \operatorname{cap}(A') \leq K\operatorname{cap}(A).$$

Among quasiconformal homeomorphisms $g : A \to A'$, the optimum and Teichmüller distance are

$$K_{\min}(A, A') := \inf_g K(g) = \max\left\{\frac{m'}{m}, \frac{m}{m'}\right\}, \qquad d_T(A, A') := \frac{1}{2}\log K_{\min}(A, A') = \frac{1}{2}\left|\log \frac{m'}{m}\right|.$$

Proof. Conformally identify A, A' with the centered round annuli below; Theorems 8.1.2 and 3.1.4, applied to essential loops modulo zero 2-modulus, give

$$\frac{1}{Km} \leq \frac{1}{m'} \leq \frac{K}{m} \iff \frac{m}{K} \leq m' \leq Km.$$

$$\lambda := \frac{m'}{m}, \qquad F_\lambda(z) := |z|^{\lambda - 1} z : A(e^{-\pi m}, e^{\pi m}) \longrightarrow A(e^{-\pi m'}, e^{\pi m'}).$$

$$\widetilde{F}_\lambda(x + iy) := \lambda x + iy, \qquad D\widetilde{F}_\lambda = \operatorname{diag}(\lambda, 1) \implies K(F_\lambda) = \max\{\lambda, \lambda^{-1}\}.$$

$$K(g) \geq \max\{\lambda, \lambda^{-1}\}, \qquad K(F_\lambda) = \max\{\lambda, \lambda^{-1}\} \implies K_{\min}(A, A') = K(F_\lambda), \quad d_T(A, A') = \frac{1}{2}|\log\lambda|.$$

The capacity identities give the remaining bounds, while the stretch proves sharpness and the distance formula. □

Remark 8.1.4 (Further directions). Natural extensions are the exact Carathéodory density and its extremizers,

$$c_A(z) := \sup_{g \in \mathrm{Hol}(A, \mathbb{D})} \frac{2|g'(z)|}{1 - |g(z)|^2} \leq \rho_A(z).$$

9. Proof of Theorem A

Theorem 9.0.1 (Modulus control). *Let $A = A(r, R)$, $A' = A(r', R')$, $m = \mathrm{mod}(A)$, $W = 2\pi m$, and $q = r/R$. Then*

(1) *For the connecting and separating curve families,*

$$\mathrm{Ext}_A(\Gamma_{\mathrm{rad}}) = m, \qquad \mathrm{Ext}_A(\Gamma_{\mathrm{sep}}) = \mathrm{cap}(A) = m^{-1}.$$

Proof.

$$A = A(r, R), \qquad m = \mathrm{mod}(A) \overset{3.1.4}{\Longrightarrow} \mathrm{Ext}_A(\Gamma_{\mathrm{rad}}) = m, \qquad \mathrm{Ext}_A(\Gamma_{\mathrm{sep}}) = \mathrm{cap}(A) = m^{-1}.$$

□

(2) *If $f \in \mathrm{Hol}(A(q,1), \mathbb{C})$, $f(z) = \sum_{n \in \mathbb{Z}} a_n z^n$, and $\sup_{A(q,1)} |f| \leq M$, then*

$$|a_n| \leq M \quad (n \geq 0), \qquad |a_{-n}| \leq M q^n \quad (n \geq 1),$$

$$\sup_{|z| = \sqrt{q}} |f(z) - a_0| \leq \frac{2M\sqrt{q}}{1 - \sqrt{q}}.$$

Proof. Put $s := \sqrt{q} = e^{-W/2}$ and $b_k := s^k a_k$. Then

$$b_k = s^k a_k \overset{4.1.2, 4.1.3}{\Longrightarrow} |b_k| \leq M s^{|k|}.$$

Hence, for $n \geq 0$ and $j \geq 1$,

$$|a_n| = s^{-n}|b_n| \leq M, \qquad |a_{-j}| = s^j |b_{-j}| \leq M s^{2j} = M q^j.$$

Moreover,

$$e^{-\pi m} = e^{-W/2} = \sqrt{q} \overset{4.1.4}{\Longrightarrow} \sup_{|z| = \sqrt{q}} |f(z) - a_0| \leq \frac{2M\sqrt{q}}{1 - \sqrt{q}}.$$

□

(3) *For $C_* := \{|z| = \sqrt{rR}\}$,*

$$\rho_A(z) = \frac{\pi}{W|z|} \csc\left(\frac{\pi}{W} \log\left(\frac{|z|}{r}\right)\right), \qquad \ell_A(C_*) = \frac{\pi}{m}.$$

Moreover, if $f \in \mathrm{Hol}(A, \mathbb{D})$ and $\rho_{\mathbb{D}}(\xi) := 2(1 - |\xi|^2)^{-1}$, then

$$\rho_{\mathbb{D}}(f(z))|f'(z)| \leq \rho_A(z), \qquad d_{\mathbb{D}}(f(z), f(\zeta)) \leq d_A(z, \zeta).$$

Proof.

$$W = 2\pi m, \qquad C_* = \{|z| = \sqrt{rR}\} \overset{5.1.2,\, 5.1.3}{\Longrightarrow} \rho_A(z) = \frac{\pi}{W|z|} \csc\left(\frac{\pi}{W} \log\left(\frac{|z|}{r}\right)\right), \quad \ell_A(C_*) = \frac{\pi}{m}.$$

$$f \in \mathrm{Hol}(A, \mathbb{D}) \overset{5.1.4}{\Longrightarrow} \rho_{\mathbb{D}}(f(z))|f'(z)| \leq \rho_A(z), \qquad d_{\mathbb{D}}(f(z), f(\zeta)) \leq d_A(z, \zeta).$$

□

(4) *If* $g \in \mathrm{Hol}(A, A')$ *is proper of signed degree* $n \in \mathbb{Z} \setminus \{0\}$*, then, in round coordinates,*

$$g(z) = c \cdot z^n, \qquad c \in \mathbb{C} \setminus \{0\}, \qquad \mathrm{mod}(A') = |n| \cdot \mathrm{mod}(A).$$

Proof.

$$g \in \mathrm{Hol}(A, A') \text{ proper}, \qquad n = \deg(g) \neq 0 \overset{6.1.3}{\Longrightarrow} g(z) = c \cdot z^n, \qquad c \neq 0, \qquad \mathrm{mod}(A') = |n| \cdot \mathrm{mod}(A).$$

□

(5) *If* $h : A \to A'$ *is a* K*-quasiconformal homeomorphism, then*

$$K^{-1} \cdot \mathrm{mod}(A) \leq \mathrm{mod}(A') \leq K \cdot \mathrm{mod}(A).$$

Proof.

$$h : A \to A' \text{ is } K\text{-quasiconformal} \overset{8.1.3}{\Longrightarrow} K^{-1} \cdot \mathrm{mod}(A) \leq \mathrm{mod}(A') \leq K \cdot \mathrm{mod}(A).$$

□

Part 2. **Appendix**

Appendix A. Foundational tools

A.1. **Cauchy bounds, coverings, and scales.**

Lemma A.1.1 (Cauchy bounds, [7, 18]). *Let $U \subset \mathbb{C}$ be a domain, $\overline{D(z_0,d)} \subset U$, and $f \in \mathrm{Hol}(U,\mathbb{C})$. Then*
$$|f^{(m)}(z_0)| \le \frac{m!}{d^m} \max_{|z-z_0|=d} |f(z)| \qquad (m \in \mathbb{Z}_{\ge 0}).$$
If $g(z) = \sum_{n\in\mathbb{Z}} a_n z^n \in \mathrm{Hol}(A(r,R),\mathbb{C})$, then
$$|a_n| \le \rho^{-n} \max_{|z|=\rho} |g(z)| \qquad (r<\rho<R,\ n \in \mathbb{Z}).$$

Proof. Cauchy's integral formulas give both estimates directly.
$$f^{(m)}(z_0) = \frac{m!}{2\pi i}\int_{|z-z_0|=d} \frac{f(z)}{(z-z_0)^{m+1}}\,dz \implies |f^{(m)}(z_0)| \le \frac{m!}{d^m}\max_{|z-z_0|=d}|f(z)|.$$
$$a_n = \frac{1}{2\pi i}\int_{|z|=\rho} \frac{g(z)}{z^{n+1}}\,dz \implies |a_n| \le \rho^{-n}\max_{|z|=\rho}|g(z)|.$$
This proves both bounds. □

Lemma A.1.2 (Periodic extraction, [7, 9]). *Let $a<b$ be real and $S_{a,b} := \{w \in \mathbb{C} : a < \Re(w) < b\}$. Then*
$$e^{w_1} = e^{w_2} \iff w_2 - w_1 \in 2\pi i\mathbb{Z},$$
and every $2\pi i$-periodic $F \in \mathrm{Hol}(S_{a,b},\mathbb{C})$ has the locally absolutely and uniformly convergent expansion
$$F(w) = \sum_{n\in\mathbb{Z}} b_n e^{nw}, \qquad b_n(x) := \frac{e^{-nx}}{2\pi}\int_0^{2\pi} F(x+iy)e^{-iny}\,dy \equiv b_n \qquad (a<x<b).$$
Moreover, if Ω is connected and $h \in \mathrm{Hol}(\Omega,\mathbb{C})$ satisfies $h(\Omega) \subset \mathbb{Z}$, then h is constant.

Proof. The fiber relation, discreteness, descent, and Laurent expansion give the claims.
$$e^{w_2-w_1} = 1, \quad w_2 - w_1 = u+iv \iff e^u = 1, \quad e^{iv} = 1 \iff u = 0, \quad v \in 2\pi\mathbb{Z}.$$
$$\Omega \text{ connected}, \quad h \text{ continuous}, \quad h(\Omega) \subset \mathbb{Z} \implies h(\Omega) \text{ connected} \implies h(\Omega) = \{k\}.$$
$$f(e^w) := F(w), \qquad e^{w_1} = e^{w_2} \implies F(w_1) = F(w_2), \qquad F = f\circ\exp, \quad f \in \mathrm{Hol}(A(e^a,e^b),\mathbb{C}).$$
$$z = e^{x+iy}, \quad dz = iz\,dy \implies \frac{1}{2\pi i}\int_{|z|=e^x}\frac{f(z)}{z^{n+1}}\,dz = \frac{e^{-nx}}{2\pi}\int_0^{2\pi} F(x+iy)e^{-iny}\,dy = b_n(x).$$
$$f(z) = \sum_{n\in\mathbb{Z}} b_n z^n, \qquad b_n(x) \equiv b_n \implies F(w) = f(e^w) = \sum_{n\in\mathbb{Z}} b_n e^{nw}.$$
The convergence is locally absolute and uniform, proving the fiber, constancy, and Fourier assertions. □

Proposition A.1.3 (Equal-modulus shells). *Let $A := A(r,R)$, $W := \log(R/r) = 2\pi\mathrm{mod}(A)$, and $N \in \mathbb{Z}_{\ge 1}$. The radii*
$$r_j := re^{jW/N} \qquad (0 \le j \le N)$$
cut A into shells $A_j := A(r_j, r_{j+1})$, $0 \le j < N$, satisfying
$$\mathrm{mod}(A_j) = \frac{\mathrm{mod}(A)}{N}.$$
Under $p_r : S_{0,W} \to A$, $p_r(w) := re^w$, these shells are equal-width cells. With $\pi_{12}(X_1,X_2,X_3) := X_1 + iX_2$, the cover has the helicoidal realization
$$\Psi_r(x+iy) := (re^x\cos y, re^x\sin y, y), \qquad \pi_{12}\circ\Psi_r = p_r, \qquad \Psi_r(w+2\pi i) = \Psi_r(w) + (0,0,2\pi).$$

Proof. Both conclusions follow by taking logarithmic radius.

$$\frac{r_{j+1}}{r_j} = e^{W/N} \implies \operatorname{mod}(A_j) = \frac{1}{2\pi}\log\frac{r_{j+1}}{r_j} = \frac{W}{2\pi N} = \frac{\operatorname{mod}(A)}{N}.$$

$$\log\frac{|p_r(x+iy)|}{r} = x, \qquad \frac{jW}{N} < x < \frac{(j+1)W}{N} \iff r_j < |p_r(x+iy)| < r_{j+1}.$$

$$\pi_{12}(\Psi_r(x+iy)) = re^x(\cos y + i\sin y) = p_r(x+iy), \qquad \Psi_r(x+i(y+2\pi)) = \Psi_r(x+iy) + (0,0,2\pi).$$

This proves the shell and helicoid formulas. $\square$

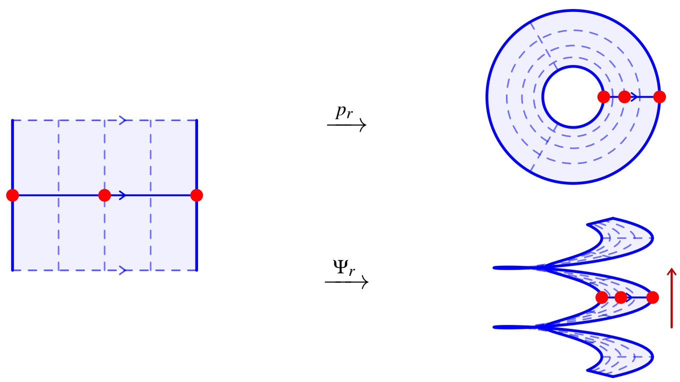

FIGURE 5. Equal-width strip cells map to equal-modulus shells under p_r and to helicoidal bands under Ψ_r; the red arrow records $y \mapsto y + 2\pi$.

Proposition A.1.4 (Spherical compactification)**.** *Inverse stereographic projection is*

$$\Sigma(z) := \frac{(2\Re(z), 2\Im(z), |z|^2 - 1)}{1+|z|^2} : \widehat{\mathbb{C}} \longrightarrow \mathbb{S}^2, \qquad \Sigma(\infty) := (0,0,1).$$

For $h_\rho := (\rho^2-1)/(\rho^2+1)$,

$$\Sigma(A(r,R)) = \{X \in \mathbb{S}^2 : h_r < X_3 < h_R\},$$

and

$$\operatorname{mod}(A(r,R)) = \frac{1}{4\pi}\log\frac{(1+h_R)(1-h_r)}{(1-h_R)(1+h_r)}.$$

In particular, $\Sigma(A(q,1)) = \{X \in \mathbb{S}^2 : h_q < X_3 < 0\}$, *and* $h_q \downarrow -1$ *as* $q \downarrow 0$.

Proof. The third coordinate records radius monotonically.

$$X_1^2 + X_2^2 = \frac{4\rho^2}{(1+\rho^2)^2}, \qquad X_3 = h_\rho, \qquad X_1^2+X_2^2+X_3^2 = 1, \qquad h'_\rho = \frac{4\rho}{(1+\rho^2)^2} > 0.$$

$$h_\rho = \frac{\rho^2-1}{\rho^2+1} \iff \rho^2 = \frac{1+h_\rho}{1-h_\rho}.$$

$$\frac{R}{r} = \left(\frac{(1+h_R)(1-h_r)}{(1-h_R)(1+h_r)}\right)^{1/2} \implies \operatorname{mod}(A(r,R)) = \frac{1}{4\pi}\log\frac{(1+h_R)(1-h_r)}{(1-h_R)(1+h_r)}.$$

This proves the band and modulus formulas. $\square$

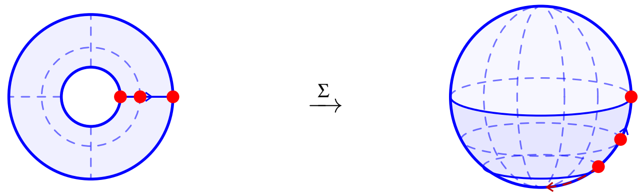

FIGURE 6. Inverse stereographic projection sends $|z| = \rho$ to $X_3 = h_\rho$, so $A(q,1)$ becomes $h_q < X_3 < 0$, with $h_q \downarrow -1$ as $q \downarrow 0$.

Proposition A.1.5 (Certified test family). *Let $A := A(q,1)$, $W := \log(1/q) = 2\pi\mathrm{mod}(A)$, $s := \sqrt{q}$, $N \in \mathbb{Z}_{\geq 0}$, $0 < \eta < 1$, and $\xi_{-N}, \dots, \xi_N \in \mathbb{C}$ not all zero. Put*

$$C := \sum_{|n| \leq N} |\xi_n|, \qquad b_n := \frac{\eta e^{-|n|W/2} \xi_n}{C}, \qquad f_N(z) := \sum_{|n| \leq N} b_n \left(\frac{z}{s}\right)^n.$$

Then $f_N \in \mathrm{Hol}(A, \mathbb{D})$, $\|f_N\|_{L^\infty(A)} \leq \eta$, and $|b_n| \leq \eta e^{-|n|W/2}$. With

$$\Delta_A(z) := \frac{W}{2} - \left|\log \frac{|z|}{s}\right|, \qquad \kappa_A(z) := \frac{1}{|z|\Delta_A(z)}, \qquad u := \frac{\pi \Delta_A(z)}{W},$$

the exact diagnostics are

$$\frac{\rho_A(z)}{\kappa_A(z)} = \frac{u}{\sin u} \in \left[1, \frac{\pi}{2}\right], \qquad Q_N(z) := \frac{2|f_N'(z)|}{(1 - |f_N(z)|^2)\rho_A(z)} \leq 1.$$

Proof. The centered coordinate certifies boundedness before any sampling.

$$z = s e^{x+iy}, \quad |x| < \frac{W}{2} \implies \left| b_n \left(\frac{z}{s}\right)^n \right| \leq \frac{\eta |\xi_n|}{C} e^{-|n|W/2 + nx} \leq \frac{\eta |\xi_n|}{C}.$$

$$|f_N(z)| \leq \frac{\eta}{C} \sum_{|n| \leq N} |\xi_n| = \eta < 1, \qquad |b_n| \leq \eta e^{-|n|W/2}.$$

$$\rho_A(z) = \frac{\pi}{W|z|} \csc\left(\frac{\pi \Delta_A(z)}{W}\right), \qquad \frac{\rho_A(z)}{\kappa_A(z)} = \frac{u}{\sin u} \overset{5.1.4}{\implies} Q_N(z) \leq 1.$$

This proves all certified bounds. ∎

For $q = e^{-2\pi m}$, the exact first-mode scale is $\delta_1(m) := e^{-\pi m} = \sqrt{q}$; the rounded benchmarks are

m	.5	1.5	2.5	3.5	4.5
$q = e^{-2\pi m}$	4.321×10^{-2}	8.070×10^{-5}	1.507×10^{-7}	2.814×10^{-10}	5.255×10^{-13}
$\delta_1(m)$	2.079×10^{-1}	8.983×10^{-3}	3.882×10^{-4}	1.678×10^{-5}	7.249×10^{-7}

TABLE 1. The inner radius and first-mode damping factor decay exponentially with modulus.

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